

Automated Loan Statement Analysis System

Designed by: MIHIR YADAV

Role: Solution Architect & Domain Expert

Built using: Python + Claude AI (Opus 4)

Manappuram Finance Ltd.

Commercial Vehicle Loan Department

Tested on: 200+ Real Loan PDFs

Time Saved: 45 min → Under 1 min per account

	A	B	C	D	E	F	G
1	Sr No	Due Date	EMI Amount	Deposit Date	Deposit Amount	DPD (Days)	Status
2	1	14-Mar-2024	23,770	15-Mar-2024	23,770	1	Paid
3	2	14-Apr-2024	23,770	14-Apr-2024	23,770	0	Paid
4	3	14-May-2024	23,770	29-May-2024	23,770	15	Paid
5	4	14-Jun-2024	23,770	14-Jun-2024	23,770	0	Paid
6	5	14-Jul-2024	23,770	14-Jul-2024	23,770	0	Paid
7	6	14-Aug-2024	23,770	14-Aug-2024	23,770	0	Paid
8	7	14-Sep-2024	23,770	14-Sep-2024	23,770	0	Paid
9	8	14-Oct-2024	23,770	14-Oct-2024	23,770	0	Paid
10	9	14-Nov-2024	23,770	30-Nov-2024	23,770	16	Paid
11	10	14-Dec-2024	23,770	14-Dec-2024	23,770	0	Paid
12	11	14-Jan-2025	23,770	14-Jan-2025	23,770	0	Paid
13	12	14-Feb-2025	23,770	14-Feb-2025	23,770	0	Paid
14	13	14-Mar-2025	23,770	26-Mar-2025	23,770	12	Paid
15	14	14-Apr-2025	23,770	14-Apr-2025	23,770	0	Paid
16	15	14-May-2025	23,770	28-May-2025	23,770	14	Paid
17	16	14-Jun-2025	23,770	14-Jun-2025	23,770	0	Paid
18	17	14-Jul-2025	23,770	14-Jul-2025	23,770	0	Paid
19	18	14-Aug-2025	23,770	30-Aug-2025	23,770	16	Paid
20	19	14-Sep-2025	23,770	14-Sep-2025	23,770	0	Paid
21	20	14-Oct-2025	23,770	22-Oct-2025	23,770	8	Paid
22	21	14-Nov-2025	23,770	14-Nov-2025	23,770	0	Paid
23	22	14-Dec-2025	23,818		0	69	Unpaid

ABOUT ME

MIHIR YADAV

Operations Executive | Manappuram Finance Ltd.

Background:

- M.Com in Accountancy and Business Statistics
- 2+ years in Commercial Vehicle Loan Operations
- Daily work involves loan disbursement, MIS reporting, delinquency tracking
- Hands-on with Excel, Power BI, Tally

My Unique Position :I understand WHAT a loan statement contains because I read them EVERY DAY.

I Know:

- What a NACH bounce looks like in a ledger
- Why a ₹500 debit is a charge, not a payment
- How advance payments should be tracked
- When a reversal cancels a receipt
- What knock-off entries mean

NO DEVELOPER KNOWS THIS WITHOUT DOMAIN GUIDANCE

THE BUSINESS PROBLEM

WHAT WE DEAL WITH DAILY: Loan Account Statement PDF

- 10-30 pages per account
- 50-200 transaction lines
- Mix of billings, receipts, charges, bounces
- Broken text formatting from PDF extraction
- Multiple date formats
- Junk entries (GST, fees, legal charges)

WHAT WE NEED TO FIND: For EACH monthly EMI

- When was EMI due?
- How much was due?
- When was it actually paid?
- How much was paid and many days late? (DPD)
- Was it Paid / Partial / Unpaid?
- Did any payment bounce?

THE PROBLEM: 45 minutes per account (manual review)

- ✗ Missed bounces (looks like a receipt)
- ✗ Junk charges confused with real payments
- ✗ Advance payments assigned to wrong month
- ✗ No standardized format for reporting
- ✗ Difficult to audit or cross-verify

11-FEB-2022	11-FEB-2022	34308003	INSTALLMENT BILLING	MFRJMFLONS000005181344-BILLING	34900.00	0.00	34900.00
12-FEB-2022	11-FEB-2022	34330984	RECEIPT	MFRJMFLONS000005181344-RECEIPT DEPOSIT	0.00	34900.00	0.00
11-MAR-2022	11-MAR-2022	34872127	INSTALLMENT BILLING	MFRJMFLONS000005181344-BILLING	34900.00	0.00	34900.00
11-MAR-2022	11-MAR-2022	34882101	RECEIPT	MFRJMFLONS000005181344-RECEIPT DEPOSIT	0.00	34900.00	0.00
11-APR-2022	11-APR-2022	35447386	INSTALLMENT BILLING	MFRJMFLONS000005181344-BILLING	34900.00	0.00	34900.00
12-APR-2022	11-APR-2022	35463441	RECEIPT	MFRJMFLONS000005181344-RECEIPT DEPOSIT	0.00	34900.00	0.00
15-APR-2022	13-APR-2022	35501205	RECEIPT	MFRJMFLONS000005181344-RECEIPT DEPOSIT (REVERSAL)	34900.00	0.00	34900.00

				LOAN ACCOUNT STATEMENT FOR MFRJMFLONS000005181344			
MANOJ YADAV				Issue Date:16/12/2025			
Transaction Date	Value Date	Trans ID	Transaction Type	Description	Debit	Credit	Balance
15-APR-2022	15-APR-2022	0	CHARGES	CHEQUE BOUNCE CHARGE-RECEIVABLE	500.00	0.00	35400.00
30-APR-2022	30-APR-2022	0	CHARGES	LATE PAYMENT INTEREST-RECEIVABLE	698.00	0.00	36098.00
04-	03-	35908031	RECEIPT	MFRJMFLONS000005181344-RECEIPT DEPOSIT	0.00	34900.00	1198.00

THE RAW PDF — WHAT WE’RE DEALING WITH

PDF STRUCTURE (16 pages):

Pages 1-2: Loan Account Details

Loan No:	MFRJMFLONS...344
Sanction Amount:	₹1168042
EMI:	₹34900/month
Tenure:	47 months
Rate:	18.11%
Period:	Sep-2021 to Jul-2025

Pages 3-4: Instruments Cleared/Bounced (cheque and NACH details)

Pages 5-7: Loan Account Transaction Details (principal + interest breakup)

→ WE SKIP THIS SECTION

Pages 8-16: Customer Transaction Details→ TARGET(the actual ledger we process)

THE BROKEN TEXT PROBLEM:

What PDF gives us:

Line 1: "11-" Line 2: "SEP-2021" Line 3: "11-" Line 4: "SEP-2021" Line 5: "31530055"

Line 6: "INSTALLMENT" Line 7: "BILLING" Line 8: "34900.00" Line 9: "0.00" Line 10: "34900.00"

What it should be:

“11-SEP-2021 11-SEP-2021 31530055 INSTALLMENT BILLING 34900.00 0.00 34900.00”

Mr MANOJ YADAV

VILLAGE NAWAB WALI POST RAGHUNATHPURA TEHSIL KOTPUTLI JAIPUR,,,,,RAJASTHAN

Pin Code - 303107

Ph - 9001773774

Email -

Loan Sanction Date	Loan Sanction Amount(Rs.)	Disbursed Amount(Rs.)	PLR / Base Rate (%)	Current Rate of Interest (%) (Interest Rate Type)	Installment Paid No. / Amount (Rs.)	Installment Pending No. / Amount (Rs.)	Future Installment No. / Amount (Rs.)
10-AUG-2021	1168042.00	1168042.00	18.11	18.11	47/1640300.00	0/	0/0.00

Agreement Id:	MFRJMFLONS000005181344	Loan Status:	Live
Product:	COMMERCIAL VEHICLES	Branch:	SHAHUPURA
Scheme:	F-USED COMMERCIAL VEHICLE	Anchor Code:	PRIME LENDING RATE
Anchor Catagory:	Base Rate	No. of Adv. Inst. / Amount:	0/RS. 0.00
Installment plan:	Equated installments/Rentals	Priority Sector Lending:	NA
Currency:	INR(INR)	Asset Classification:	REGULAR
Instl. Start Date:	11-SEP-2021	Repayment Frequency:	Monthly
Instl. End Date:	11-JUL-2025	Disbursal Status:	Fully Disbursed
Tenure:	47 Months	Disbursal Payment Status:	Disbursal Paid
Collateral linked Loan(s)		Is Legally:	Nil
LPI Rate(%)/Amount	18% Per Annum		
Is Settlement:	Nil		
Repo Marked/Unmark:	Repo Not Marked		
Repo Marked date:			
REPO RELEASE DATE:			

Current Overdue (Rs.)	Current Late Payment Interest (Rs.)	Current LPF Receivable (Rs.)	Interest Accrued At Termination (Rs.)	Balance Principal (Rs.)	Total Outstanding (Rs.)
0.00		1135.00	0.00	0.00	9301.88

11- SEP-2021	11- SEP-2021	31530055	INSTALLMENT BILLING	MFRJMFLONS000005181344-BILLING	34900.00	0.00	34900.00
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MY SOLUTION APPROACH

MY ROLE (Domain Architect):

- Identified the business problem
- Designed complete 6-stage pipeline
- Defined ALL business rules & edge cases
- Specified what counts as billing/receipt/bounce
- Tested output against 200+ real PDFs
- Identified errors and directed corrections
- Documented the workflow

AI's ROLE (Code Developer):

- Wrote Python code based on my specifications
- Implemented regex patterns I describe
- Built functions I designed
- Fixed bugs I identified during testing

Tool Used: Claude AI (Opus 4)

THE DOMAIN KNOWLEDGE MADE THIS PROJECT WORK.
AI JUST TRANSLATED MY KNOWLEDGE INTO CODE.

SYSTEM ARCHITECTURE

(6-STAGE PROCESSING PIPELINE)

Stage 1: PDF INGESTION & LEDGER DETECTION

- Open PDF with pdfplumber
- Scan each page for trigger keyword
- "CUSTOMER TRANSACTION DETAILS" found
- Skip pages 1-7, process pages 8-16

STAGE 2: BROKEN DATE RECONSTRUCTION

- Detect 6 types of broken date patterns
- Merge split lines back into complete txns
- "11-" + "SEP-2021" → "11-SEP-2021"
- Handle fused dates: "11-SEP2021"

STAGE 3: TRANSACTION CLASSIFICATION

- Extract dates, amounts, description
- Classify: BILLING / RECEIPT / BOUNCE / JUNK
- Apply 15+ keyword filters
- Skip KNOCK-OFF, DISBURSAL, BOOKING entries

STAGE 4: BOUNCE DETECTION & RECEIPT CLEANING

- Match each bounce to its original receipt
- Cancel same-day reversal and deposit pairs
- Remove bounced receipts from clean list
- Detect foreclosure payments

STAGE 5: RECEIPT-TO-BILL ASSIGNMENT

- Window-based assignment (date proximity)
- Cumulative EMI tracking for advance detect
- Multiple receipts → single bill aggregation
- Deposit date = EMI threshold crossing date

STAGE 6: DPD CALCULATION & EXCEL OUTPUT

- DPD = Deposit Date - Due Date
- Status: Paid / Partial / Unpaid / Foreclosed
- Color-coded formatted Excel report
- Frozen headers, date formatting

STAGE 1: PDF INGESTION — HOW IT WORKS

PDF FILE (16 PAGES)

pdfplumber.open(pdf)
For each page: extract_text()

CHECK 1: FORECLOSURE?

Look for:
"LOAN CLOSURE"

"FORECLOSURE"

"LOAN STATUS: CLOSED"

CHECK 2: LEDGER START?

Look for:
"CUSTOMER TRANSACTION DETAILS"

Page 1-7: NOT found- → **SKIP page**

Page 8: FOUND! → **START PROCESSING**

THE PDF HAS TWO TRANSACTION SECTIONS:

- ❌ "Loan Account Transaction Details" (Pages 5-7)
 - Principal + Interest breakup
 - NOT what we need
 - Would give wrong data if parsed

Loan Account Transaction Details							
Transaction Date	Value Date	Trans ID	Transaction Type	Description	Debit	Credit	Balance
10-AUG-2021	10-AUG-2021	30917356	Loan Booking / Disbursal	MFRJMFLONS000005181344-Loan Booking	1168042.00	0.00	1168042.00
11-SEP-2021	11-SEP-2021	31530055	Installment Billing	MFRJMFLONS000005181344-Billing	0.00	16354.00	1151688.00
11-OCT-2021	11-OCT-2021	32073315	Installment Billing	MFRJMFLONS000005181344-Billing	0.00	17756.00	1133932.00
11-NOV-2021	11-NOV-2021	32610668	Installment Billing	MFRJMFLONS000005181344-Billing	0.00	17458.00	1116474.00

- ✅ "Customer Transaction Details" (Pages 8-16)
 - Actual billing, receipt, bounce entries
 - THIS is our target data

Customer Transaction Details							
Transaction Date	Value Date	Trans ID	Transaction Type	Description	Debit	Credit	Balance
11-SEP-2021	11-SEP-2021	31530055	INSTALLMENT BILLING	MFRJMFLONS000005181344-BILLING	34900.00	0.00	34900.00
13-SEP-2021	11-SEP-2021	31550303	RECEIPT	MFRJMFLONS000005181344-RECEIPT DEPOSIT	0.00	34900.00	0.00
11-OCT-2021	11-OCT-2021	32073315	INSTALLMENT BILLING	MFRJMFLONS000005181344-BILLING	34900.00	0.00	34900.00
13-OCT-2021	11-OCT-2021	32096292	RECEIPT	MFRJMFLONS000005181344-RECEIPT DEPOSIT	0.00	34900.00	0.00

STAGE 2: BROKEN DATE RECONSTRUCTION

PROBLEM: PDF text extraction splits dates randomly
AFTER REBUILD: 9 LINES → 1 CLEAN TRANSACTION LINE

PATTERN 1: DAY-ONLY START

Broken: "11-"
"SEP-2021"

Fixed: "11-SEP-2021"

PATTERN 2: DAY-MONTH WITHOUT YEAR

Broken: "11-SEP-"
"2021"

Fixed: "11-SEP-2021"

PATTERN 3: TWO PARTIAL DATES ON SAME LINE

Broken: "11-SEP- 11-SEP-"
"2021 2021"

Fixed: "11-SEP-2021 11-SEP-2021"

PATTERN 4: FUSED MONTH-YEAR (NO HYPHEN)

Broken: "11-SEP2021"

Fixed: "11-SEP-2021"

PATTERN 5: FULL DATE + PARTIAL DATE

Broken: "11-SEP-2021 13-SEP-"
"2021"

Fixed: "11-SEP-2021 13-SEP-2021"

PATTERN 6: SPLIT ACROSS 3+ LINES

Broken: "11-"
"13-SEP-"
"SEP-"
"2021"
"2021"

Fixed: "11-SEP-2021 13-SEP-2021"

ANALYTICS RELEVANCE:

This is a DATA QUALITY problem. In any analyst role, cleaning messy source data is 60-80% of the work. This stage handles the ETL (Extract, Transform, Load) challenge that every data analyst faces daily.

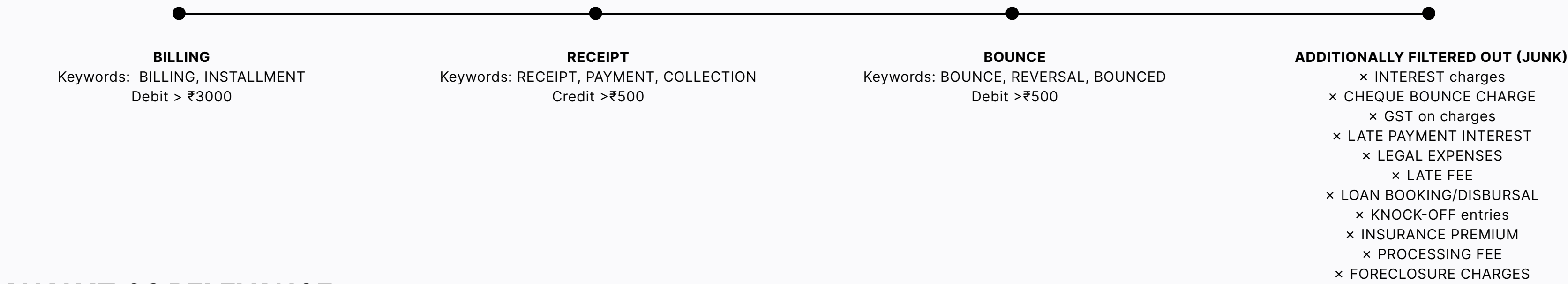
STAGE 3: TRANSACTION CLASSIFICATION

WHAT EACH LINE CONTAINS:

"11-SEP-2021 11-SEP-2021 31530055 INSTALLMENT BILLING MFRJMFLONS000005181344-BILLING 34,900.00 0.00 34,900.00"

EXTRACTED:

Transaction Date: 11-SEP-2021
Value Date: 11-SEP-2021
Description: INSTALLMENT BILLING
Debit: ₹34,900
Credit: ₹0
Balance: ₹34,900



ANALYTICS RELEVANCE:

This is essentially a CLASSIFICATION problem categorizing unstructured text data into business categories using rule-based decision trees. Similar to what analysts do with transaction categorization in banking & credit risk.

STAGE 4: BOUNCE DETECTION & RECEIPT CLEANING

IN A LOAN LEDGER, A BOUNCED PAYMENT APPEARS AS:

DAY 1: RECEIPT (CREDIT ₹34,900) ← LOOKS PAID
DAY 3: REVERSAL (DEBIT ₹34,900) ← PAYMENT FAILED
DAY 20: RECEIPT (CREDIT ₹34,900) ← ACTUAL PAYMENT

WE MUST: REMOVE THE BOUNCED RECEIPT (DAY 1)
KEEP THE REVERSAL AS BOUNCE RECORD
KEEP THE REPLACEMENT PAYMENT (DAY 20)

```
Stats: 47 Bills | 52 Receipts | 11 Bounces | 0 Foreclosure Receipts
[Bounce] Removed receipt amt=34,900 date=11-APR-2022 matched bounce amt=34,900 on 13-APR-2022
[Bounce] No matching receipt for bounce amt=500 on 15-APR-2022 - skipping
[Bounce] No matching receipt for bounce amt=500 on 13-MAY-2022 - skipping
[Bounce] Removed receipt amt=34,900 date=11-MAY-2022 matched bounce amt=34,900 on 13-MAY-2022
[Bounce] No matching receipt for bounce amt=500 on 13-JUL-2022 - skipping
[Bounce] Removed receipt amt=34,900 date=11-JUL-2022 matched bounce amt=34,900 on 13-JUL-2022
[Bounce] No matching receipt for bounce amt=500 on 13-DEC-2022 - skipping
[Bounce] Removed receipt amt=34,900 date=11-DEC-2022 matched bounce amt=34,900 on 13-DEC-2022
[Bounce] No matching receipt for bounce amt=500 on 13-FEB-2023 - skipping
[Bounce] Removed receipt amt=34,900 date=11-FEB-2023 matched bounce amt=34,900 on 13-FEB-2023
[Bounce] Removed receipt amt=34,900 date=11-JAN-2025 matched bounce amt=34,900 on 13-JAN-2025
```

3 TYPES OF BOUNCES DETECTED:

RECEIPT REVERSAL

Text: "RECEIPT DEPOSIT (REVERSAL)"
Pattern: Credit entry followed by Debit reversal

COLLECTION BOUNCED

Text: "COLLECTION BOUNCED"
Pattern: Direct bounce notification, Debit entry

NACH/CHEQUE BOUNCE

Text: "Bounced because Funds Insufficient"
Pattern: No matching receipt (NACH auto-debit)

MATCHING ALGORITHM:

For each BOUNCE entry, find matching RECEIPT:
Criteria: 1. Amount difference < ₹1
2. Receipt date ≤ Bounce date
3. Date gap ≤ 15 days
4. Receipt not already consumed by another bounce

SAME-DAY CANCELLATION: If REVERSAL + DEPOSIT on same date and same amount
→ Both entries cancel out completely
→ Neither receipt nor bounce counted

RESULT:

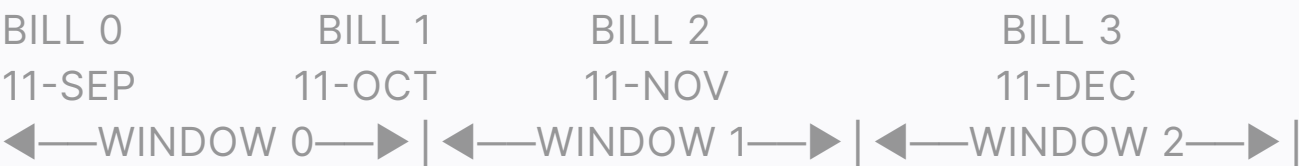
Raw Receipts: ~55
Pair Cancellations: -2
Bounce Removals: -6
Clean Receipts: ~47
= Exactly matches 47 billing months

ANALYTICS RELEVANCE:

This is a DATA RECONCILIATION problem matching debits to credits, identifying failed transactions, and producing clean data. Core skill for financial analysts.

STAGE 5: RECEIPT-TO-BILL ASSIGNMENT ENGINE (The Core Algorithm)

CONCEPT: EACH BILL CREATES A TIME WINDOW



RECEIPT ON 13-SEP → FALLS IN WINDOW 0
RECEIPT ON 03-OCT → FALLS IN WINDOW 0
RECEIPT ON 15-OCT → FALLS IN WINDOW 1

NORMAL VS ADVANCE PAYMENT:

BILL	EMI	CUM EMI	CUM RECEIPTS	DECISION
0	34,900	34,900	0	→ NORMAL
1	34,900	69,800	34,900	→ NORMAL
2	34,900	104,700	69,800	→ NORMAL

ADVANCE SCENARIO:
If cum_receipts (69,800) ≥ cum_emi (69,800)
AND new receipt ≥ 50% of EMI
→ This is ADVANCE → assign to NEXT bill

ANALYTICS RELEVANCE:

This is essentially a WATERFALL ALLOCATION model distributing payments across obligations using cumulative tracking. Same concept used in revenue recognition, cash flow allocation, revenue recognition, cash flow allocation,

```
--- Receipt Assignment ---
NORMAL: 11-SEP-2021 amt=34,900 -> Bill [0] 11-SEP-2021 (cum_rcpt=0 cum_emi=34,900)
NORMAL: 11-OCT-2021 amt=34,900 -> Bill [1] 11-OCT-2021 (cum_rcpt=34,900 cum_emi=69,800)
NORMAL: 11-NOV-2021 amt=34,900 -> Bill [2] 11-NOV-2021 (cum_rcpt=69,800 cum_emi=104,700)
NORMAL: 11-DEC-2021 amt=34,900 -> Bill [3] 11-DEC-2021 (cum_rcpt=104,700 cum_emi=139,600)
NORMAL: 11-JAN-2022 amt=34,900 -> Bill [4] 11-JAN-2022 (cum_rcpt=139,600 cum_emi=174,500)
NORMAL: 11-FEB-2022 amt=34,900 -> Bill [5] 11-FEB-2022 (cum_rcpt=174,500 cum_emi=209,400)
NORMAL: 11-MAR-2022 amt=34,900 -> Bill [6] 11-MAR-2022 (cum_rcpt=209,400 cum_emi=244,300)
NORMAL: 03-MAY-2022 amt=34,900 -> Bill [7] 11-APR-2022 (cum_rcpt=244,300 cum_emi=279,200)
NORMAL: 30-MAY-2022 amt=34,900 -> Bill [8] 11-MAY-2022 (cum_rcpt=279,200 cum_emi=314,100)
NORMAL: 11-JUN-2022 amt=34,900 -> Bill [9] 11-JUN-2022 (cum_rcpt=314,100 cum_emi=349,000)
NORMAL: 31-JUL-2022 amt=34,900 -> Bill [10] 11-JUL-2022 (cum_rcpt=349,000 cum_emi=383,900)
NORMAL: 11-AUG-2022 amt=34,900 -> Bill [11] 11-AUG-2022 (cum_rcpt=383,900 cum_emi=418,800)
```

STAGE 6: DPD CALCULATION & EXCEL OUTPUT

FORMULA: $DPD = \text{DEPOSIT DATE} - \text{DUE DATE (IN DAYS)}$

RULES:

- $DPD < 0$ (paid early) $\rightarrow DPD = 0$
- $DPD = 0$ (paid on time) $\rightarrow DPD = 0$
- $DPD > 0$ (paid late) $\rightarrow DPD = \text{actual days}$
- Unpaid bill $\rightarrow DPD = \text{Today} - \text{Due Date}$

STATUS CLASSIFICATION:

Deposit $\geq$ EMI $\rightarrow$ "Paid" 

$0 < \text{Deposit} < \text{EMI} \rightarrow$ "Partial" 

Deposit = 0 $\rightarrow$ "Unpaid" 

Foreclosure amount $\rightarrow$ "Foreclosed" 

SAFETY CHECKS APPLIED:

- DATE FILL: If bill has no receipt, use next available receipt date for DPD estimation
- MONOTONICITY: If $\text{Bill}[n] \text{ deposit_date} > \text{Bill}[n+1] \text{ deposit_date} \rightarrow$ fix backward (earlier bills can't logically pay later)
- FORECLOSURE: Last receipt $> 1.5 \times \text{avg EMI} \rightarrow$ Flagged as foreclosure, separate row

EXCEL OUTPUT FORMAT:

Sr No | Due Date | EMI Amt | Dep Date | DPD(Days) | Status |

WHY DPD MATTERS (CREDIT RISK CONTEXT):

DPD RANGE	CLASSIFICATION
0 DAYS	CURRENT (ON-TIME)
1-30 DAYS	SMA-0 (SPECIAL MENTION)
31-60 DAYS	SMA-1
61-90 DAYS	SMA-2
91+ DAYS	NPA (NON-PERFORMING ASSET)

This classification directly impacts:

- Loan provisioning requirements
- Credit score of borrower
- Collection priority and strategy
- Regulatory reporting (RBI compliance)

	A	B	C	D	E	F	G
	Sr No	Due Date	EMI Amount	Deposit Date	Deposit Amount	DPD (Days)	Status
1							
2	1	11-Sep-2021	34,900	11-Sep-2021	34,900	0	Paid
3	2	11-Oct-2021	34,900	11-Oct-2021	34,900	0	Paid
4	3	11-Nov-2021	34,900	11-Nov-2021	34,900	0	Paid
5	4	11-Dec-2021	34,900	11-Dec-2021	34,900	0	Paid
6	5	11-Jan-2022	34,900	11-Jan-2022	34,900	0	Paid
7	6	11-Feb-2022	34,900	11-Feb-2022	34,900	0	Paid
8	7	11-Mar-2022	34,900	11-Mar-2022	34,900	0	Paid
9	8	11-Apr-2022	34,900	03-May-2022	34,900	22	Paid
10	9	11-May-2022	34,900	30-May-2022	34,900	19	Paid
11	10	11-Jun-2022	34,900	11-Jun-2022	34,900	0	Paid
12	11	11-Jul-2022	34,900	31-Jul-2022	34,900	20	Paid
13	12	11-Aug-2022	34,900	11-Aug-2022	34,900	0	Paid
14	13	11-Sep-2022	34,900	11-Sep-2022	34,900	0	Paid
15	14	11-Oct-2022	34,900	11-Oct-2022	34,900	0	Paid
16	15	11-Nov-2022	34,900	11-Nov-2022	34,900	0	Paid
17	16	11-Dec-2022	34,900	31-Dec-2022	34,900	20	Paid
18	17	11-Jan-2023	34,900	11-Jan-2023	34,900	0	Paid
19	18	11-Feb-2023	34,900	28-Feb-2023	34,900	17	Paid
20	19	11-Mar-2023	34,900	11-Mar-2023	34,900	0	Paid
21	20	11-Apr-2023	34,900	11-Apr-2023	34,900	0	Paid
22	21	11-May-2023	34,900	11-May-2023	34,900	0	Paid
23	22	11-Jun-2023	34,900	11-Jun-2023	34,900	0	Paid

EDGE CASES HANDLED BY THE ENGINE

These edge cases were identified from my domain experience and specified to the AI during development:

EDGE CASE 1: SAME-DAY REVERSAL + DEPOSIT

04-Aug-2022: Receipt ₹34,900 (deposit)

04-Aug-2022: Receipt ₹34,900 (reversal)

Problem: Both appear as separate transactions

Solution: Cancel both (net effect = zero)

EDGE CASE 2: DOUBLE BOUNCE IN SAME MONTH

Dec-2022:

11-Dec: Receipt ₹34,900 → bounced 13-Dec

31-Dec: Receipt ₹34,900 → bounced 02-Jan!

02-Jan: Receipt ₹34,900 → finally cleared

Problem: 3 receipts, 2 bounces, 1 actual payment

Solution: Match each bounce to nearest receipt, keep only the final successful payment

EDGE CASE 3: NACH BOUNCE WITH NO RECEIPT

Aug-2023: "Bounced because Funds Insufficient"

Problem: NACH auto-debit failed, but there's no corresponding receipt entry (bank rejected it)

Solution: Record as bounce, no receipt to remove

EDGE CASE 4: BOUNCE CHARGE vs REAL BOUNCE

"CHEQUE BOUNCE CHARGE-RECEIVABLE ₹500.00"

"GST ON BOUNCE CHARGE RECEIVABLE ₹76.27"

Problem: Contains "BOUNCE" keyword but it's a fee

Solution: Filter by amount (< ₹500 threshold)

EDGE CASE 5: KNOCK-OFF ENTRIES

"RECTIFICATION OF RECEIPT ENTRY KNOCK OFF WITH LEGAL RECEIVABLE ₹34,900"

Problem: Shows ₹34,900 credit — looks like payment

Solution: Skip entirely (internal accounting adj)

EDGE CASE 6: ADVANCE PAYMENT DETECTION

Sep-2023: Two receipts in same window

14-Sep: ₹34,900 (NEFT)

15-Sep: ₹34,900 (NACH)

Problem: Both fall in Sep window

Without advance logic: Sep = ₹69,800, Oct = ₹0

With advance logic: Sep = ₹34,900, Oct = ₹34,900

EDGE CASE 7: REFUND HANDLING

Borrower overpays → lender issues refund

Problem: Refund looks like a new debit transaction

Solution: Match refund to original receipt, remove the refunded receipt from clean list

EDGE CASE 8: FORECLOSURE DETECTION

Large final payment > 1.5x average EMI

Problem: Foreclosure payment is NOT a regular EMI

Solution: Flag separately, don't assign to any bill Show as "Foreclosed" status in orange row

EDGE CASE 9: BOOKING / DISBURSAL ENTRY

"LOAN BOOKING ₹11,68,042"

Problem: Huge credit entry, could be mistaken for a receipt

Solution: HIGH PRIORITY JUNK filter catches this before normal classification

RESULTS & BUSINESS IMPACT

BEFORE (Manual Process):

- 🕒 Time per account: 45 minutes
- ❌ Human errors: Frequent
- ❌ Missed bounces: Common
- ❌ DPD accuracy: Approximate
- ❌ Standardization: None
- ❌ Audit trail: Manual notes
- ❌ Scalability: 1 person = 10 accounts/day



AFTER (Automated Engine):

- 🕒 Time per account: Under 1 minute
- ✅ Human errors: Zero (rule-based)
- ✅ Bounce detection: 100% automated
- ✅ DPD accuracy: Exact (date math)
- ✅ Standardization: Uniform Excel format
- ✅ Audit trail: Terminal logs
- ✅ Scalability: 1 person = 400+ accounts

POTENTIAL SCALE (If deployed):

Manappuram Finance has ~4,000 branches
If each branch reviews 50 accounts/month:
Manual: 50 × 45 min = 37.5 hours/branch/month
Automated: 50 × 1 min = 50 min/branch/month
Time saved per branch: ~36 hours/month
Company-wide (4000 branches): 144,000 hours/month
That's 900 full-time employee's worth of time
saved EVERY MONTH.

Metric	Before	After
Time/account	45 min	<1 min
Time saved		98%
Accounts/day	10	400+
Error rate	5-10%	0%
Bounce detection	Manual	Auto
Report format	Varied	Standard

USE CASES & BROADER APPLICATIONS

USE CASE 1: COLLECTIONS TEAM

Problem: Need to identify delinquent accounts

How: Run engine on all accounts → sort by DPD

Benefit: Instantly see which accounts need attention

DPD > 30 → Urgent call

DPD > 60 → Field visit

DPD > 90 → Legal notice

USE CASE 2: CREDIT UNDERWRITING

Problem: Assess borrower repayment history

How: Parse existing loan statements for new loan

Benefit: Quick DPD history check before approval

Bounce rate > 20% → Higher interest rate

Max DPD > 60 → Reject or lower amount

USE CASE 3: AUDIT & COMPLIANCE

Problem: Verify DPD calculations for RBI reporting

How: Cross-check system DPD with engine output

Benefit: Independent verification tool

Catches system errors in NPA classification

USE CASE 4: PORTFOLIO ANALYTICS

Problem: Analyze payment patterns across portfolio

How: Process 100+ PDFs → aggregate DPD data

Benefit: Portfolio-level insights

- Average DPD by product type
- Bounce rate trends by month
- Seasonal payment patterns
- Early warning signals for NPA

USE CASE 5: MANAGEMENT REPORTING

Problem: Branch performance on collections

How: Process all branch accounts monthly

Benefit: Standardized collection efficiency report

- % accounts with DPD = 0
- % accounts with DPD > 30
- Branch-wise bounce rate comparison

USE CASE 6: BORROWER SELF-SERVICE

Problem: Borrower disputes DPD reported to CIBIL

How: Parse statement → show exact dates and proof

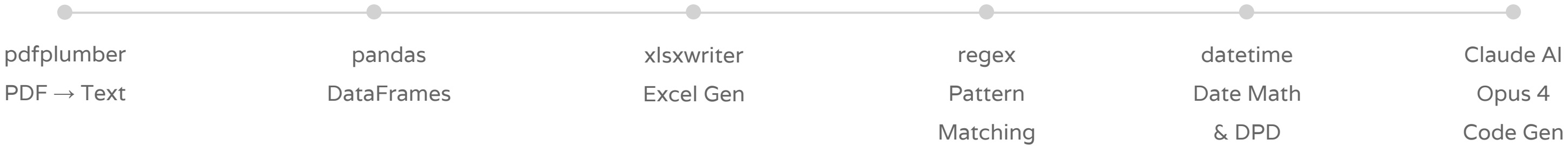
Benefit: Quick resolution with documentary evidence

The CONCEPT (parse PDF → classify → track → calculate DPD → report) is universal.

Only the business rules change per domain.

TECHNOLOGY & SKILLS DEMONSTRATED

TECHNOLOGY STACK:



SKILLS DEMONSTRATED (RELEVANT TO ANALYST ROLES):

DATA ANALYST SKILLS:

- ✓ Data extraction from unstructured sources (PDF)
- ✓ Data cleaning & validation (broken dates, junk filtering, deduplication)
- ✓ Data classification (rule-based categorization)
- ✓ Data reconciliation (bounce-receipt matching)
- ✓ ETL pipeline design (Extract, Transform, Load)
- ✓ Report generation (formatted Excel output)
- ✓ Pattern recognition in transactional data

FINANCIAL ANALYST SKILLS:

- ✓ DPD calculation & credit risk assessment
- ✓ Payment behavior analysis
- ✓ Bounce rate analysis
- ✓ Loan portfolio monitoring
- ✓ NPA/SMA classification understanding
- ✓ RBI regulatory framework awareness
- ✓ Collection efficiency metrics

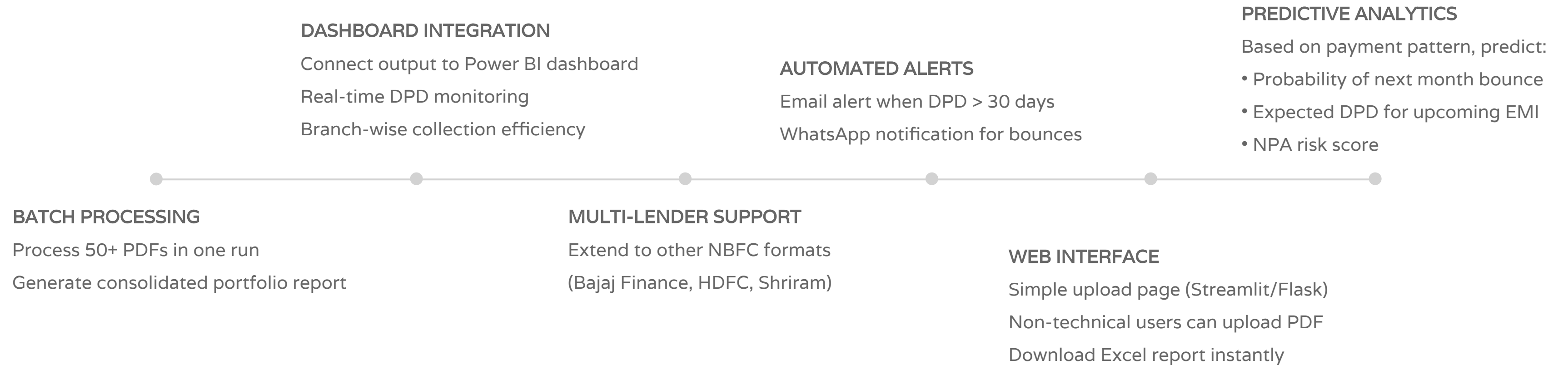
AI-ASSISTED DEVELOPMENT SKILLS:

- ✓ Problem decomposition for AI
- ✓ Business logic specification
- ✓ Prompt engineering for code generation
- ✓ Iterative testing & refinement
- ✓ Edge case identification from domain knowledge
- ✓ AI tool management (directing, not just using)

PROCESS AUTOMATION SKILLS:

- ✓ Identifying automation opportunities
- ✓ Workflow design & documentation
- ✓ Testing & validation methodology
- ✓ Impact measurement (before/after metrics)

FUTURE ENHANCEMENTS (Roadmap)



KEY TAKEAWAYS

1. DOMAIN EXPERTISE IS IRREPLACEABLE

AI wrote the code.

But AI didn't know:

- What a NACH bounce looks like
- Which ₹500 entry is a charge vs payment
- How advance payments work
- What knock-off entries mean

That knowledge came from 2+ years of reading loan statements EVERY DAY.

2. AI IS A FORCE MULTIPLIER

Without AI: I had the idea but no coding skill to build it.

With AI: I designed → AI coded → I tested
→ We iterated → Working product in weeks.

Domain Expert + AI = 10x Productivity

4. MEASURABLE IMPACT MATTERS

45 minutes → Under 1 minute

25+ PDFs tested with accurate results

Always quantify your impact. Numbers speak louder than descriptions.

3. PROCESS AUTOMATION IS ACCESSIBLE

You don't need a CS degree to automate.

You need:

- Deep understanding of the problem
- Ability to break it into logical steps
- Willingness to test and iterate
- AI tools to handle the coding

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